Indian Institute of Technology Bombay

PH 544 - Problem set 2 **General relativity** **Date 1-02-2026**

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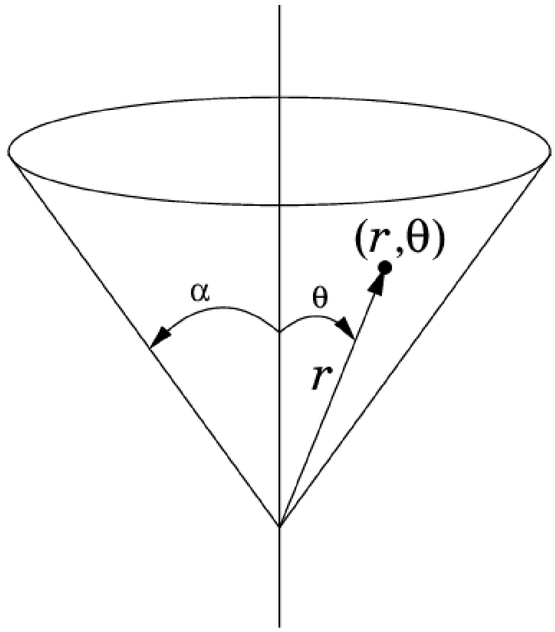
1. Consider a right cirular cone with half opening angle α.

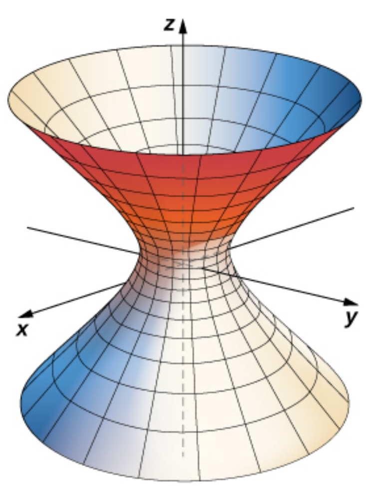


   (a) Determine the 2D metric that describes the surface of the cone.

   (b) Compute the Christoffel symbols.

   (c) Using the Euler-Lagrange equation, obtain the equations of motion.

2. A hyperboloid is a surface of revolution which can be obtained by rotating a hyperbola about the z-axis.



It is defined by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \tag{1}$$

The parametric form is given by

$$x = a \cosh v \cos\theta, \quad y = b \cosh v \sin\theta, \quad z = c \sinh v \tag{2}$$

(a) Determine the 2D metric that describes the surface of the hyperboloid.

(b) Compute the Christoffel symbols.

(c) Using the Euler-Lagrange equation, obtain the equations of motion.

3. A theoretician claims that a physical system should be described in a 3-dimensional space with the coordinates x, y, z and the metric to be:

$$ds^2 = dx^2 + dy^2 + dz^2 - \left(\frac{3}{13}dx + \frac{4}{13}dy + \frac{12}{13}dz\right)^2 \tag{3}$$

With a series of probes, experimentalist shows that it is really a two-dimensional space, and that the metric is given by:

$$ds^2 = d\zeta^2 + d\xi^2 \tag{4}$$

Who is right? If experimentalist is right, what is the transformation between (x, y, z) and (ζ, ξ)

4. Show that the line-element

$$ds^2 = g_{11}(x, y)dx^2 + 2\, g_{12}(x, y)\, dx\, dy + g_{22}(x, y)\, dy^2 \tag{5}$$

can be written as

$$ds^2 = \left(\sqrt{g_{11}}\, dx + \frac{g_{12}}{\sqrt{g_{11}}}\, dy\right)^2 + \left(g_{22} - \frac{g_{12}^2}{g_{11}}\right) dy^2 \tag{6}$$

Under what condition (6) can be written as

$$ds^2 = d\zeta^2 + d\xi^2 \tag{7}$$

5. In Geometry, you have learnt that to any smooth curve, it is possible to draw a straight line tangent to a point. Let us consider an arbitrary 2-dimensional surface given by the following line-element:

$$ds^2 = g_{11}(x, y)dx^2 + 2\, g_{12}(x, y)\, dx\, dy + g_{22}(x, y)\, dy^2 \tag{8}$$

Show that the it is always possible to match any arbitrary region of such a space by a flat space (provided the region is taken small enough).

6. In the class, we calculated first and second fundamental form for a 2-Sphere. Calculate second fundamental form and the Curvature for

(a) Cylinder

(b) Cone

(c) Hyperboloid